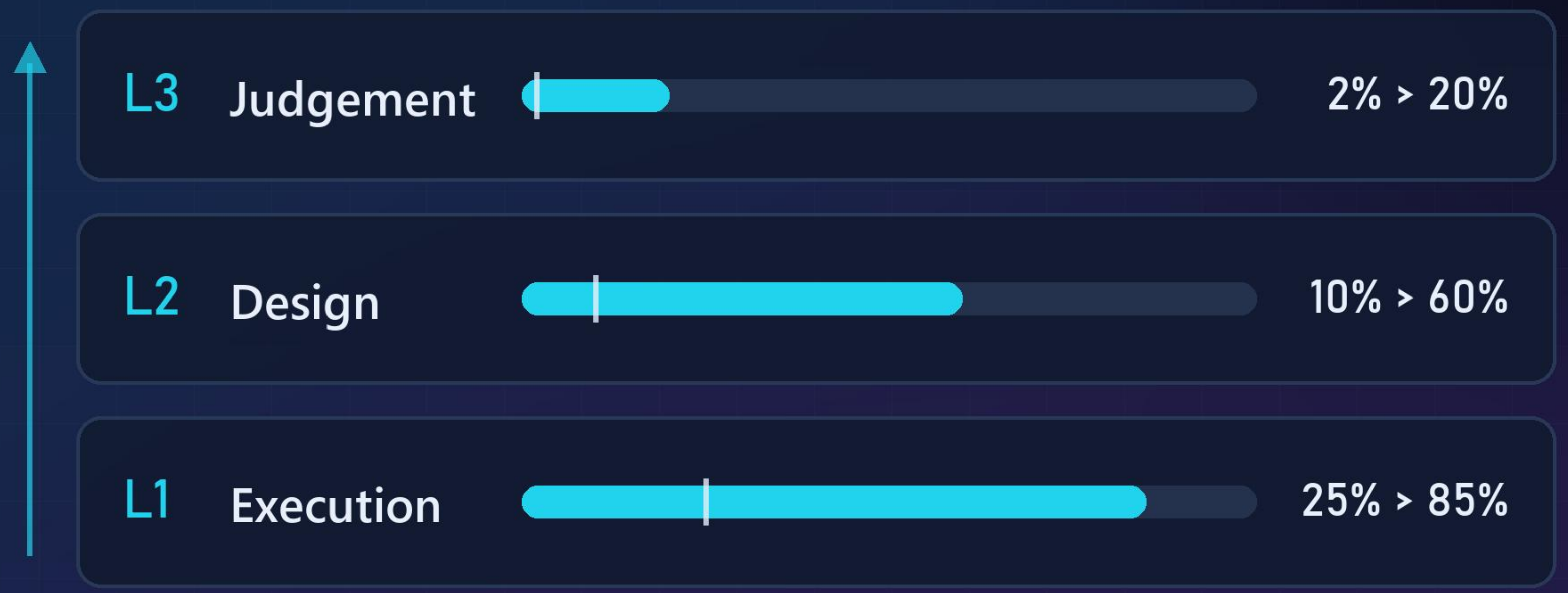
POST-SILICON VALIDATION

Who will actually do the work?

A three-layer model of engineering work, and an honest look at how fast AI can move through it.

> AI absorbs this stack from the bottom up. Layer 1 goes first because it is repetitive and well defined. Layer 3 barely moves, because its constraints are legal and physical rather than technical.



Bar shows roughly ten years out, marker shows today.
Illustrative projection. Method and limits on page 07.

THE MODEL

Every engineering job has three layers

L3 Judgement

today

2% > 20%

What matters. What is trustworthy. What ships.

Risk calls, escalation, customer commitments, lot sign-off.

L2 Design

today

10% > 60%

Which corners. Which limits. Which method.

Test strategy, limit setting, coverage decisions, trade-offs.

L1 Execution

today

25% > 85%

Writing the pattern. Grepping the datalog. Running the sweep.

Pattern coding, log parsing, sweep execution, report assembly.

AI starts at Layer 1 and works its way up. The same people stay in the flow, they just move higher in it.

THE FLOW

16 stages, one closed learning loop

Every stage holds all three layers, and every stage keeps the same ten engineers. Agents add capacity beneath them. The people move up.

SPEED

The stages do not move at the same rate

FAST

70 - 85%
at ~ +10 years

- Wafer / die screening
- Characterization analysis
- Test program development
- Debug and failure triage
- Root-cause analysis
- HVM screening and SPC
- Field monitoring

Pure data and code. No physical gate, no re-qualification gate.

MEDIUM

40 - 60%
at ~ +10 years

- Structural and scan / DFT
- Memory and peripheral validation
- Production test release
- Pilot run and yield ramp

Gated by tool qualification and correlation runs.

SLOW

10 - 30%
at ~ +10 years

- Bring-up planning
- Power-on and electrical
- Functional bring-up
- Corrective action
- Validation closure and sign-off

Physical lab work, cross-org negotiation, legal accountability.

Weighted across all 16 stages, at roughly ten years out

About 54% for a leading team. Industry median closer to 25 - 35%.

WHAT SLOWS IT DOWN

Five brakes that are not about model capability

01 **Qualification, not capability**

Changing a production test program means re-correlation, re-qualification and often customer approval. The blocker is not that the output is bad. It is that the tool which generated it is not qualified.

02 **Validation is not software**

Probe cards, load boards, thermal chambers, handlers, bench instruments. Several stages need hands. Lab robotics moves at hardware speed, not model speed.

03 **Data debt is the real 90%**

Agents need clean, consistent, queryable test data. Most organisations have siloed STDF, test names that differ per product, and the decisive context living in engineers' heads.

04 **IP sensitivity caps model quality**

Test programs and debug data are among the most guarded assets in the company. On-prem and air-gapped deployment trails frontier capability.

05 **The verification tax**

Reviewing unreliable AI output can be slower than doing the work yourself. Adoption only accelerates past a reliability threshold, and that threshold is high when a wafer lot is at stake.

WHAT SPEEDS IT UP

Three forces already in motion

Vendors will drag the industry along

AI-assisted design, verification and test-analytics tooling from the major EDA and ATE suppliers means companies do not have to build this themselves. Vendor-delivered capability adopts far faster than internal projects.

Domain LLMs already have a head start

Domain-adapted language models for chip design work have been public since 2023. That track has three years on it, and the tooling around it is maturing fast.

Adaptive test is already normal

Real-time outlier detection and dynamic part-average testing run in high-volume production today. In several flows the automation baseline is already higher than people outside test assume.

PRODUCT CLASS MATTERS MORE THAN COMPANY NAME

Memory and HVM	Fastest. Huge repetition and data volume, thin margins forcing quick ROI.
Datacenter and consumer logic	Fast. Shorter qualification chains, strong internal AI capability.
Mobile	Medium. High volume, but heavy variant and certification load.
Safety-critical automotive and industrial	Slowest in production. But pre-production analytics move at full speed, no re-qual.

THE CLAIM

What I would actually defend

Within about ten years, leading teams will have automated 40 to 60% of routine post-silicon execution, concentrated almost entirely in data analysis, triage and code generation. Physical bring-up and sign-off will barely move. Industry-wide it will be closer to 30%.

LIMITATIONS

- The dominant unknown is frontier model capability. If it plateaus, only the fast tier happens. If scaling continues, the medium tier compresses.
- The slow tier is slow for legal and physical reasons, not technical ones. More capable models will not move it much.
- The entry path is the open question. Juniors traditionally learn in Layer 1. If that layer shrinks, how the next generation is trained is unresolved.
- These are structural estimates from public industry dynamics, not internal data from any company.
- The percentages are illustrative. The ordering is the argument, not the decimals.

WHY THIS IS GOOD NEWS

- Semiconductor content keeps growing. Chiplets, 3D stacking and variant explosion are adding validation surface every year.
- Automating 60% of today's tasks is fully consistent with flat or growing validation headcount.
- The scarce skill becomes knowing when an agent is confidently wrong. That is domain expertise, and it does not transfer in from outside the industry.